FINAL PROJECT PROPOSAL

PREDICTING PATIENT SATISFACTION FROM HOSPITAL REVIEWS USING TRADITIONAL MACHINE LEARNING

IE7275 Data Mining in Engineering | Wiktoria Lasek

Problem Setting

Access to healthcare in Canada remains a significant challenge. Millions of Canadians are unable to see a doctor in a timely manner, and when they do receive care, the quality of their experience varies widely across hospitals and clinics. Patients frequently share their experiences through online reviews, providing hospitals with a valuable but largely underutilized source of feedback. Healthcare administrators and hospital management teams receive hundreds or thousands of these reviews but lack the time and tools to analyze them systematically at scale (Greaves et al., 2013).

The inability to process patient feedback efficiently means that recurring issues - such as long wait times, poor communication, or inadequate staff responsiveness - often go unaddressed. Automated tools that can classify and analyze patient sentiment would enable healthcare providers to identify problem areas quickly, prioritize improvements, and ultimately deliver better patient care. Data mining techniques offer a practical and scalable approach to this challenge, enabling pattern recognition across large volumes of unstructured text data (Manning et al., 2008).

Problem Definition

This project aims to build an automated system that reads hospital patient reviews and predicts whether the experience was positive or negative. Using traditional machine learning classification techniques, the goal is to identify which model best captures patient sentiment from text - and which words and phrases most strongly indicate satisfaction or dissatisfaction. The outcome is a practical tool that healthcare providers can use to automatically monitor patient feedback at scale, identify recurring issues, and make data-driven decisions to improve the quality of care.

Data Sources

The primary dataset used in this project is the Hospital Patient Reviews Dataset, publicly available on Kaggle (Hospital Reviews Dataset, 2023, https://www.kaggle.com/datasets/junaid6731/hospital-reviews-dataset). The dataset contains real patient reviews of hospitals along with sentiment labels and star ratings. It is freely available, contains no personally identifiable information, and raises no ethical concerns.

Data Description

Dataset name	Hospital Patient Reviews Dataset
Source	Kaggle
Number of rows	996 patients reviews
Number of columns	3 columns
Variable names	Feedback (text), Sentiment Label (0/1),Ratings (1-5)
Label distribution	728 positive (73%), 268 negative (27%)
Missing values	0 – dataset is clean
Train/Test split	80% training, 20% test
Validation strategy	5-fold cross validation

The Feedback column contains free-text patient reviews which will be converted into numerical features using TF-IDF vectorization. The Sentiment Label column serves as the target variable for binary classification. Given the class imbalance (73% positive, 27% negative), F1-score will be used as the primary evaluation metric rather than accuracy alone.

Methodology

This project follows the complete data mining process across five stages:

1. Data Exploration and Visualization: The dataset will be explored to understand the overall structure, patterns, and distribution of patient feedback. Visualizations will be generated to communicate key findings and guide modelling decisions.

2. Data Preprocessing and Feature Engineering: The review text will be cleaned through tokenization, stop-word removal, lowercasing, and punctuation removal. TF-IDF vectorization will then convert the cleaned text into numerical feature vectors suitable for machine learning models.

3. Dimension Reduction and Feature Selection: TF-IDF naturally produces high-dimensional feature spaces. The top 1,000 most important terms will be selected using TF-IDF max_features, followed by chi-squared feature selection to further reduce to the 500 most statistically significant terms - reducing noise and improving model efficiency (Manning et al., 2008).

4. Model Exploration and Selection: Five classification algorithms will be implemented and compared: Logistic Regression, Naive Bayes, Decision Tree, Support Vector Machine (Joachims, 1998), and K-Nearest Neighbors. Each model will be trained and validated using 5-fold cross validation - testing the model on different subsets of the data five times and averaging results for reliable performance estimates. This approach is especially important given the relatively small dataset size of 996 reviews.

5. Performance Evaluation and Visualization: All models will be evaluated using Accuracy, Precision, Recall, and F1-Score - with F1-Score as the primary metric given the class imbalance. Results will be visualized through comparison bar charts, confusion matrices, and ROC curves to clearly communicate which model performs best and why.

References

Greaves, F., Ramirez-Cano, D., Millett, C., Darzi, A., & Donaldson, L. (2013). Harnessing the cloud of patient experience: Using social media to detect poor quality healthcare. *BMJ Quality & Safety, 22*(3), 251-255. https://doi.org/10.1136/bmjqs-2012-001527

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Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. *European Conference on Machine Learning*. https://doi.org/10.1007/BFb0026683

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